

Institute of Actuaries of India

Subject CT7 – Business Economics

May 2014 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution : 1.5 Mark to each answer

1. D
2. B
3. C
4. D
5. C
6. A
7. C
8. C
9. D
10. C
11. A
12. B
13. C
14. B
15. D
16. A
17. D
18. B
19. B
20. D
21. C
22. B
23. B or B&D:
24. A
25. D
26. B
27. B
28. B
29. C
30. A

[Q.No. 1 to 30=45 Marks]

Solution 31 :

These can take the form of:

- maximum prices (ceilings) – to protect consumers, eg rent controls
- minimum prices (floors) – to protect producers, to protect consumers against themselves in the case of alcohol (!); to protect workers' wages.

Price control might have following disadvantages:

- Ceilings below the equilibrium price will cause shortages. Non-price rationing needs to be introduced and underground (black) markets might develop.

- Floors above the equilibrium price will lead to surpluses, eg wine lakes, unemployed workers. In the case of goods, governments either have to set quotas for firms or they need to buy the surplus supplies, store them, and then either destroy them or sell them cheaply on the world market.

Further problems of government intervention:

- problems caused by measurement difficulties and a lack of information
- bureaucracy and inefficiency
- a negative effect on incentives
- the reduction in individual freedom of choice
- the adverse effect on efficiency if there are frequent changes in government policy.

[6 Marks]

Solution 32 :

- Crowding out is a phenomena where increased public expenditure diverts money or resources away from the private sector.
- If the government relies on pure fiscal policy that means it does not finance an increase in budget deficit by increasing the money supply. Then it will have to borrow the money from non-banking private sector.
- It will face competition from private sector for finance and will thus have to offer higher interest rates. This will force the private sector also to offer higher interest rates.
- These increased interest rates will discourage investment by firms and individuals from buying on credit. This will reduce private sector consumption/investment.
- Hence, a rise in government expenditure may be completely offset by a fall in consumption and investment with the result that aggregate demand does not rise at all.

[5 Marks]

Solution 33 :

Because water is essential to life, the TU received from water exceeds the TU received from gold. However, the price we are willing to pay for each unit of a commodity depends not on the TU but on the MU. We consume so much water that the MU of the last unit of water consumed is very low. Therefore, we are willing to pay only a very low price for the last unit of water consumed. Since all units of water consumed are identical we pay the same low price on all other units of water consumed. On the other hand, we purchase so few gold objects that the MU of the last gold object purchased is very high. Therefore, we are willing to pay a high price for this last gold object and for all the other gold items purchased.

(It is important that MU instead of TU being the driving factor is understood)

Points mentioning the relative supply of these resources

[3 Marks]

Solution 34 :

- a) For $x > 50$, both firms have a dominant strategy.
- b) The unique NE is (*Low, Low*).
- c) For $40 < x < 50$ only Firm 2 has a dominant strategy.
- d) The unique NE in this case is still (*Low, Low*).
- e) For $x < 40$, there are zero dominant strategies,
- f) No NE exists.

[6 Marks]**Solution 35 :**

The effect on money supply would depend on who is buying the treasury bills and bonds.

- If the central bank buys them, the effect on the government's account is neutral (i.e. money comes in and then it goes out when it is spent) but when the government spends its money in the economy, this spending increases deposits in the banks. These can be used as a basis for credit creation. This credit creation will result in increased money supply.
- If the banks buy Treasury bills, the money supply increases. Treasury bills are liquid assets and can be used as a basis for credit creation. But in case the banks buy gilts they cannot be used for credit creation. The government can minimize the effect on the money supply by issuing gilts.
- If the non-bank private sector buys bonds or bills, then the money supply will be unchanged. The money is withdrawn from banks, given to the government, and then re-deposited in banks as the government spends the money i.e. the money simply changes hands.

[6 Marks]**Solution 36 :**

- a) Expected Salary = 40 K in option 1 and in Option 2, it is $60K \cdot 0.5 + 20K \cdot 0.5 = 40 K$

[1]

- b) We have to calculate the Expected utility under each options.

Option 1 = $2 \cdot 40 + 10 \cdot (40)^{(0.5)} = 82,000$

Option 2, it is $\{2 \cdot 60 + 10 \cdot (60)^{(0.5)}\} \cdot 0.5 + \{2 \cdot 20 + 10 \cdot (20)^{(0.5)}\} \cdot 0.5 = 81,932$.

So, option 1 would put him in a better off situation.

[3]

- c) The two offers have the same expected value. Since Mr Desperate prefers the certain salary to the risky salary, it follows that he is risk averse. Notice that utility allows us to add preference to the raw numbers. Expected value is a raw salary here, but expected utility allows us to compare two equal outcomes by including preference (utility) in the equation.

[2]**[Total Marks-6]**

Solution 37 :

- a) Suppose the firm is required to produce Q units of output. Show how the cost minimizing quantity of labor depends on the quantity Q. Show how the cost minimizing quantity of capital depends on quantity Q.

Starting with the condition we have $MP_L/MP_K = w/r$

$$\Rightarrow (L^{1/2} + K^{1/2}) * L^{-1/2} / (L^{1/2} + K^{1/2})K^{-1/2} = 2/1$$

$$\Rightarrow K/L = 4$$

$$\Rightarrow K = 4L$$

$$\text{Then } Q = (L^{1/2} + (4L)^{1/2})^2$$

$$= (3L^{1/2})^2$$

$$= 9L$$

$$\text{Then } L = Q/9$$

Inserting this back into the solution for K above gives

$$K = 4Q/9 \quad [2]$$

- b) Find the equation of the firm's long-run total cost curve.

$$TC = 2(Q/9) + (4Q/9)$$

$$= 2Q/3$$

[1]

- c) Find the equation of the firm's long-run average cost curve.

$$AC = TC/Q = 2/3$$

[1]

- d) Find the solution to the firm's short-run cost-minimization problem when capital is fixed at a quantity of 9 units (i.e. K=9)

When $Q \leq 9$ the firm needs no labor. If $Q > 9$ the firm must hire labor. Setting

$K = 9$ and plugging in for capital in the production function yields

$$Q = (L^{1/2} + K^{1/2})^2$$

$$Q^{1/2} = L^{1/2} + K^{1/2}$$

$$= L^{1/2} + 9^{1/2} = L^{1/2} + 3$$

$$\Rightarrow L^{1/2} = Q^{1/2} - 3$$

$$\Rightarrow L = (Q^{1/2} - 3)^2$$

$$\text{So, } L = (Q^{1/2} - 3)^2 \text{ if } Q > 9$$

$$\text{Else } 0 \text{ if } Q \leq 9$$

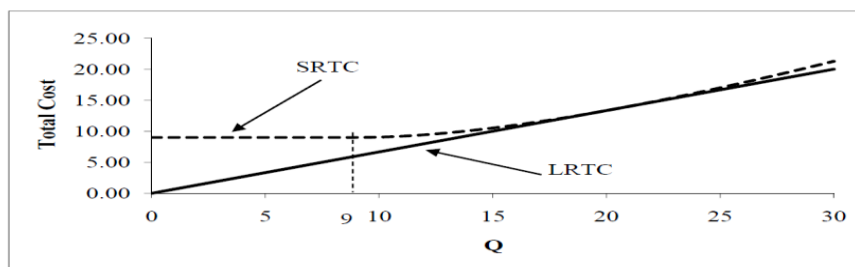
[3]

- e) Find the short-run total cost curve, and graph it along with the long-run total cost curve.

$$TC = 2(Q^{1/2} - 3)^2 + 9 \text{ when } Q > 9$$

$$= 9 \text{ when } Q \leq 9$$

Graphically, short-run and long-run total cost are shown in the following figure.



[3]

f) Find the associated short-run average cost curve

$$\begin{aligned} AC &= TC/Q \\ &= \{(2(Q^{1/2}-3)^2 + 9)\} / Q \text{ if } Q > 9 \\ &= 9/Q \text{ if } Q \leq 9 \end{aligned}$$

[1]

[Total Marks-11]

Solution 38 :

- a) In industries operating under cost conditions (such as constant returns to scale) that make the existence of perfect competition feasible, the dissolution of a monopoly (by government antitrust action) into a large number of perfectly competitive firms will result in a greater long-run equilibrium output for the industry, a lower commodity price, and usually a lower LAC than under monopoly. However, because of cost and technological conditions, it is not desirable to break up a natural monopoly into a large number of perfectly competitive firms. In dealing with natural monopolies, the government usually chooses to regulate them rather than break them up.
- b) There is a great deal of disagreement on whether monopoly or perfect competition leads to more technological progress. Since a monopolist usually makes long-run profits while perfect competitors do not, the monopolist has more resources to devote to research and development (R& D). The monopolist is also more likely to retain the benefits of the technological advance it introduces. A technological advance introduced by a perfect competitor which leads to lower costs and short-run profits is easily and quickly copied by other firms, and this eliminates the profits of the firm that introduced it. On the other hand, a monopolist may feel very secure in its position and have no incentive to engage in research and development and to innovate

[6 Marks]

Solution 39 :

- a) The natural barriers to entry into oligopolistic industries like the automobile, aluminum, and steel industries are the smallness of the markets in relation to efficient operation and the huge amounts of capital and specialized inputs required to start efficient operation. Some artificial barriers to entry are control over raw materials, patents, and government franchise. When entry is blocked or at least restricted, the firms in an oligopolistic industry can earn long-run profits.

[2]

- b) In the long run, oligopoly may lead to the following harmful effects: (1) $P > MC$ and so there is an under allocation of the economy's resources to the firms in the oligopolistic industry; (2) the oligopolist usually does not produce at the lowest point on its LAC curve; and (3) when oligopolists produce a differentiated product, too much may be spent on advertising and model changes.

For technological reasons, many products (such as automobiles, steel, etc.) cannot be produced under conditions of perfect competition (because their cost of production would be prohibitively high). In addition, oligopolists spend a great deal of their profits on research and development, and this may lead to faster technological advance and a higher standard of living than if the industry were organized along more competitive lines. Finally, some advertising is useful since it informs customers, and some product differentiation has the economic value of satisfying the different tastes of different consumers.

[4]**[Total Marks-6]**
